



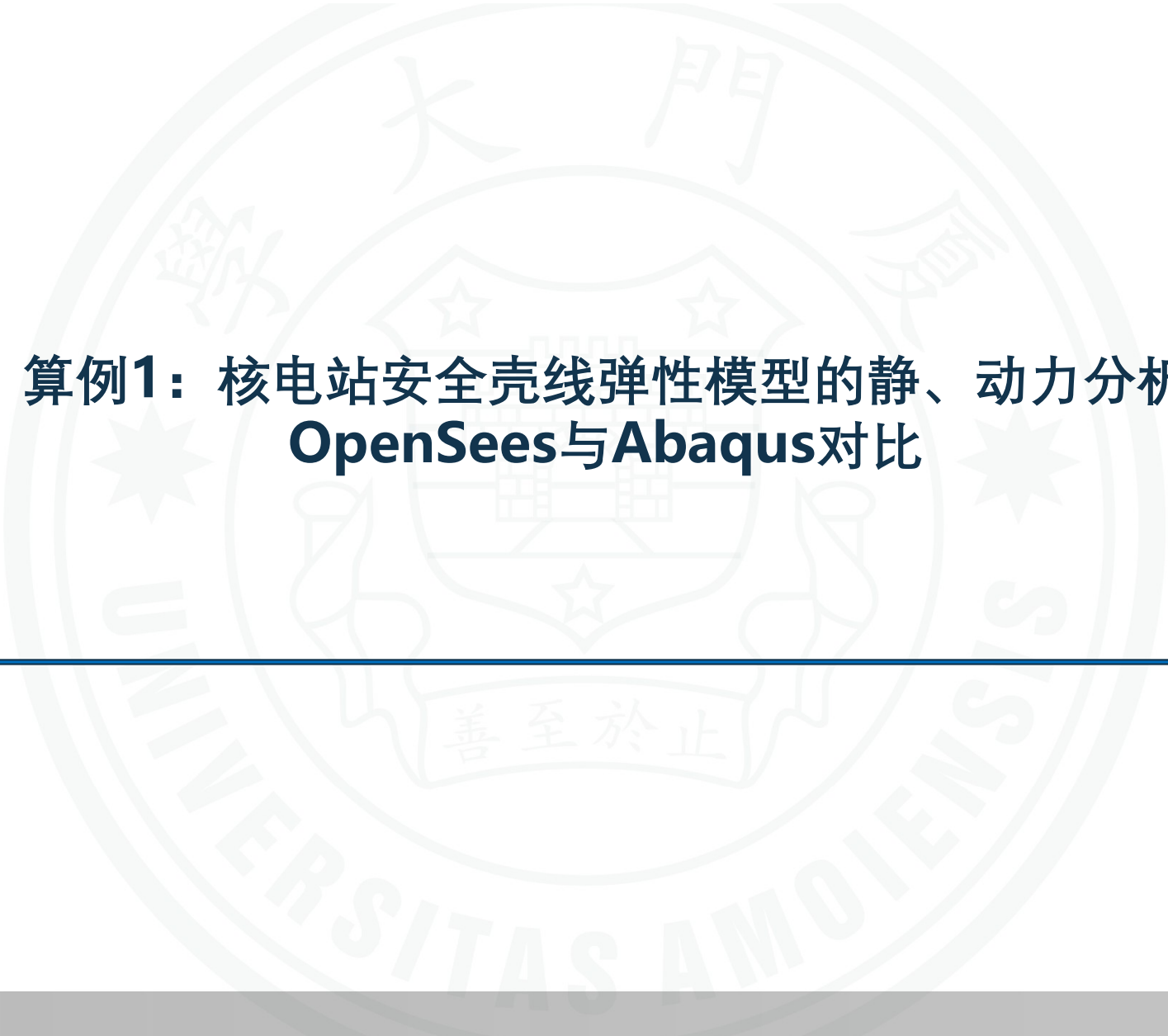
廈門大學
XIAMEN UNIVERSITY

OpenSees性能評估 Performance assessment of OpenSees

古泉 (Quan Gu)

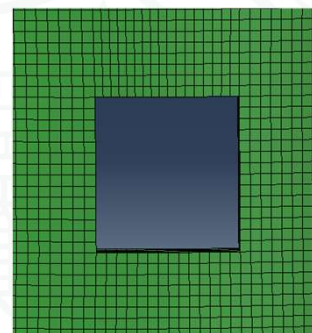
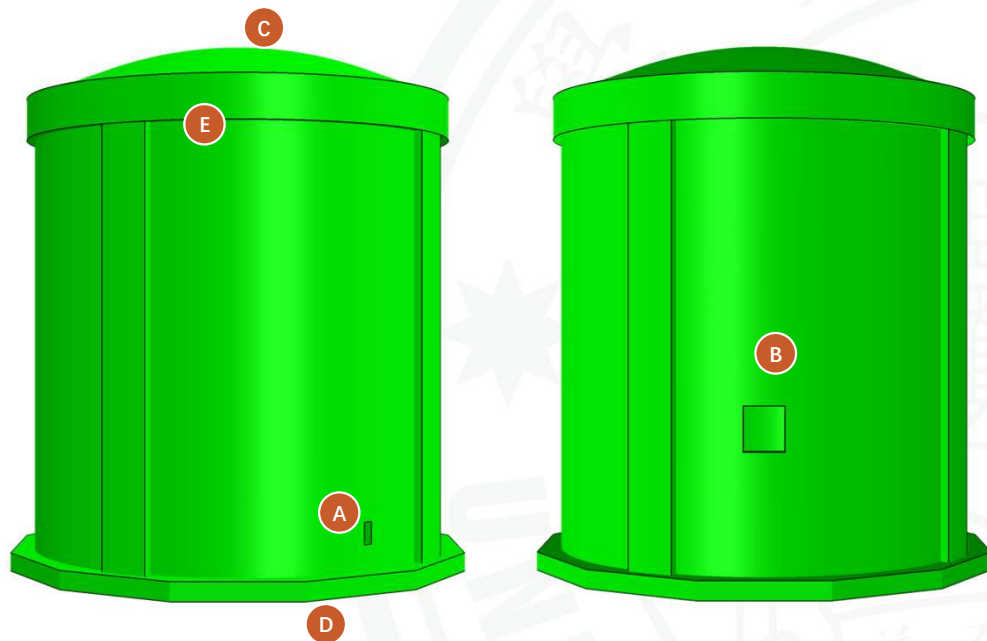
林哲 (Zhe Lin)

2026

The background of the slide features a large, faint watermark of the Amo University seal. The seal is circular, with the university's name in Chinese characters '廈門大學' at the top and 'AMOIENSIS' at the bottom. The central part of the seal contains a shield with a building and stars, and a banner at the bottom with the motto '止於至善'.

算例1：核电站安全壳线弹性模型的静、动力分析 OpenSees与Abaqus对比

1. Abaqus 竖向模型视图与节点布置

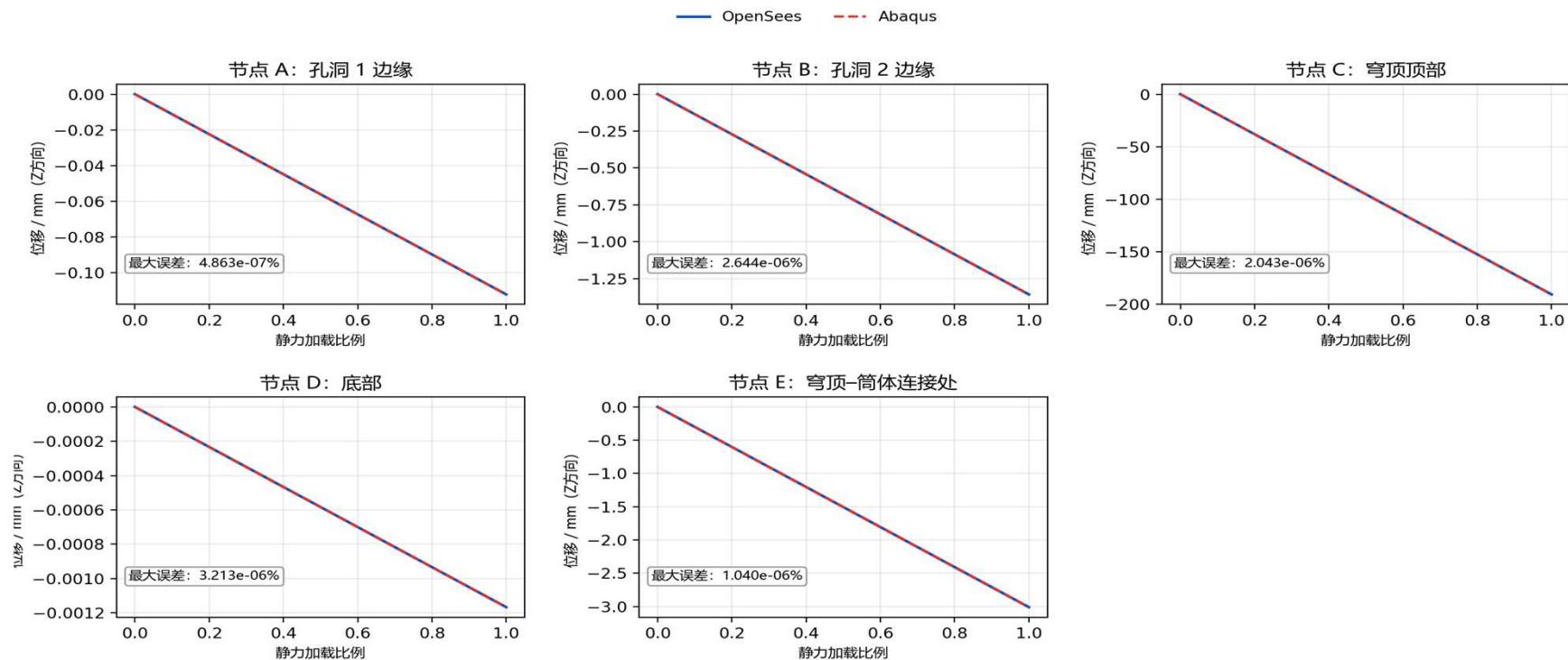


五个对比节点

- A 孔洞 1 边缘
- B 孔洞 2 边缘
- C 穹顶顶部
- D 底部
- E 穹顶-筒体连接处

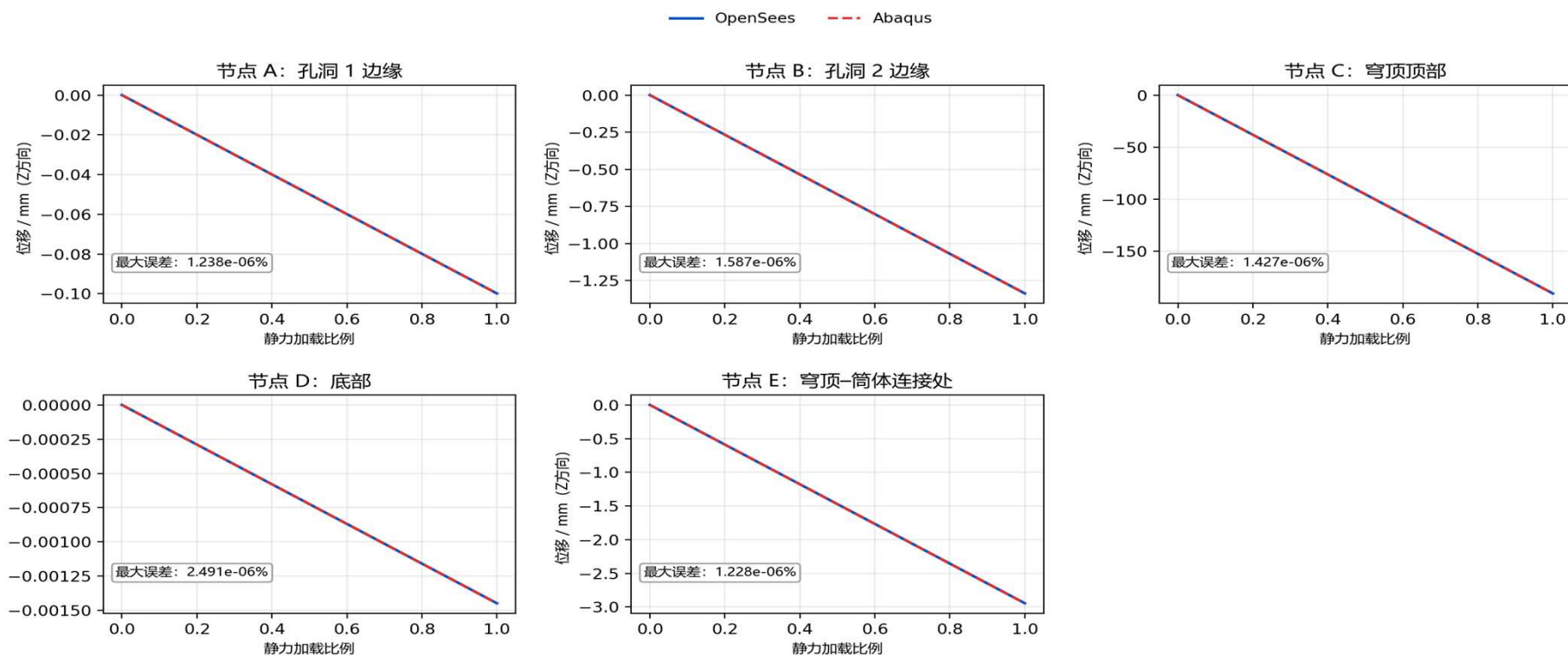
节点	706,755
单元	557,297
自由度	1,671,891
材料参数	线弹性 $E=29600$ MPa、 $\nu=0.15$

2. 自重 + 节点集中力 静力分析-OpenSees与Abaqus位移对比 (z方向)



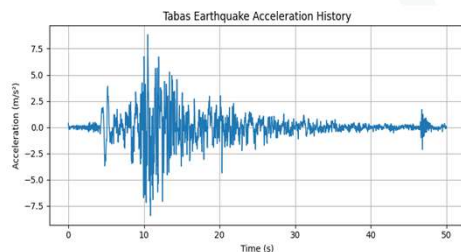
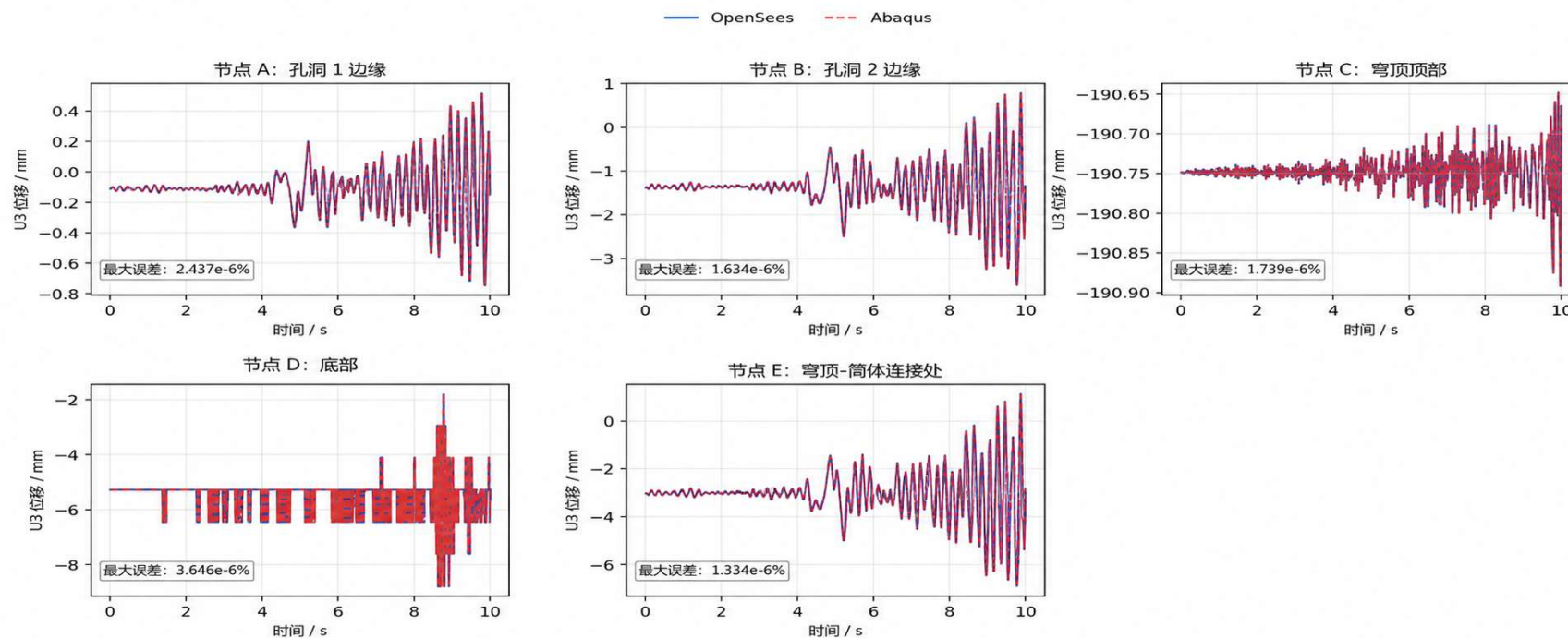
	Abaqus	OpenSees
计算时间 (单步)	约 5.89 s	约 6.0 s (第一步之后)
最大误差	最大相对误差约 3.21e-6 %	

3. 自重 + 5 kPa 内压 静力分析-OpenSees与Abaqus位移对比 (z方向)



	Abaqus	OpenSees
计算时间 (单步)	约 6.22 s	约 6.33 s (第一步之后)
最大误差	最大相对误差约 2.49e-6 %	

4. 地震作用下动力分析-OpenSees与Abaqus位移对比 (z方向)



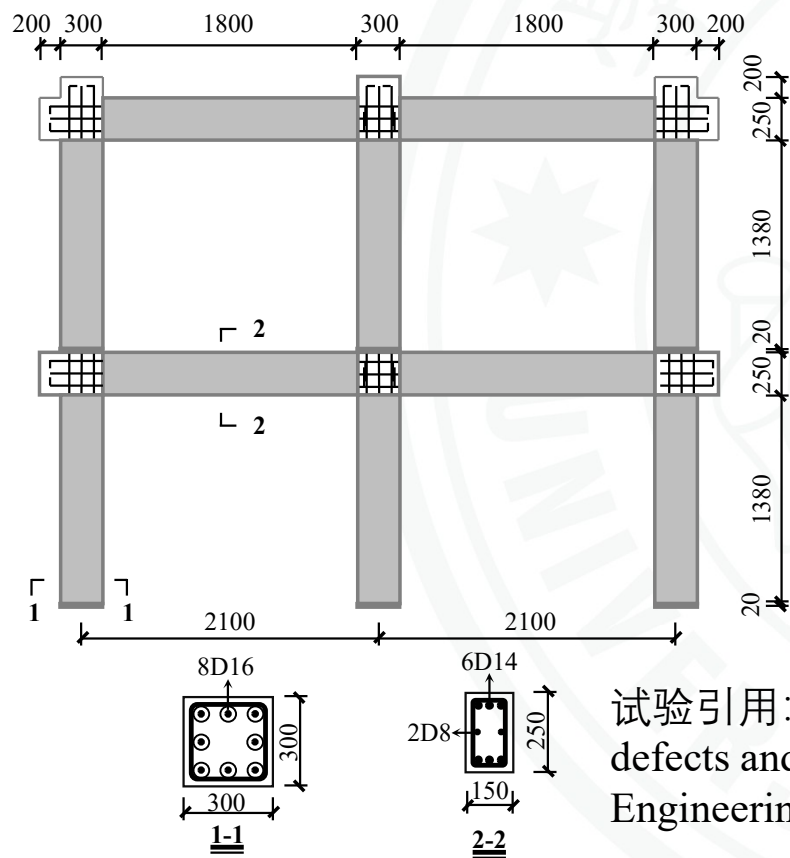
	Abaqus	OpenSees
计算时间 (500 步)	2 h 23 min	2 h 26 min
最大误差	最大相对误差约 $3.646e-6\%$	

The background of the slide features a large, light gray watermark of the Tsinghua University seal. The seal is circular, with the university's name in Chinese characters '清華大學' at the top and 'TSINGHUA UNIVERSITY' at the bottom. In the center is a shield containing a building and stars.

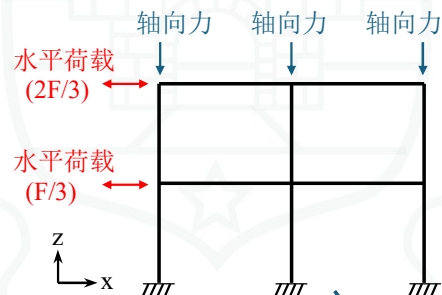
算例2：装配式非线性RC框架的OpenSees纤维 梁柱模拟与试验对比

1. 结构概况

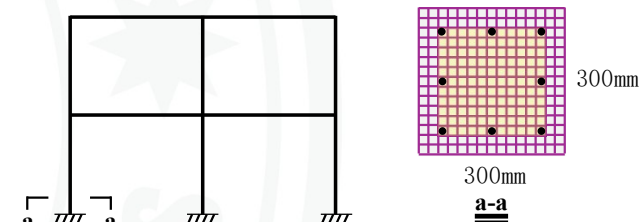
几何尺寸及配筋



加载方式及缺陷位置



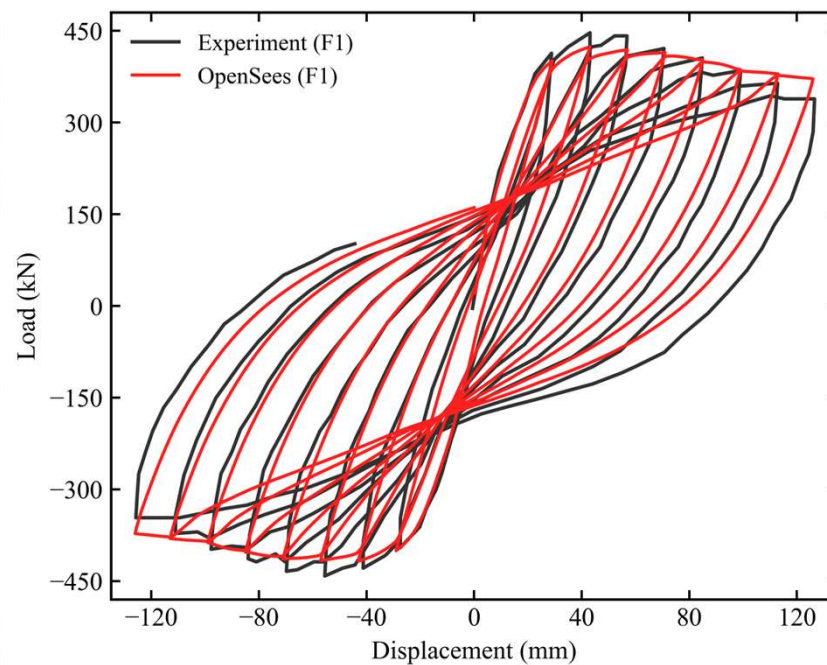
F1纤维模型



水平荷载 F 为两层施加荷载之和
水平位移取第二层梁的实测位移

试验引用: Li X, Xiao S, Gao R, Harries K A, Wang Z, Xu Q. Effects of grout sleeve defects and their repair on the seismic performance of precast concrete frame structures[J]. Engineering Structures, 2021, 242: 112619. DOI: 10.1016/j.engstruct.2021.112619.

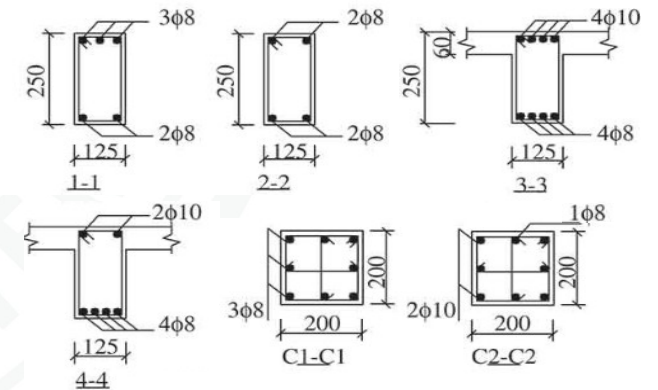
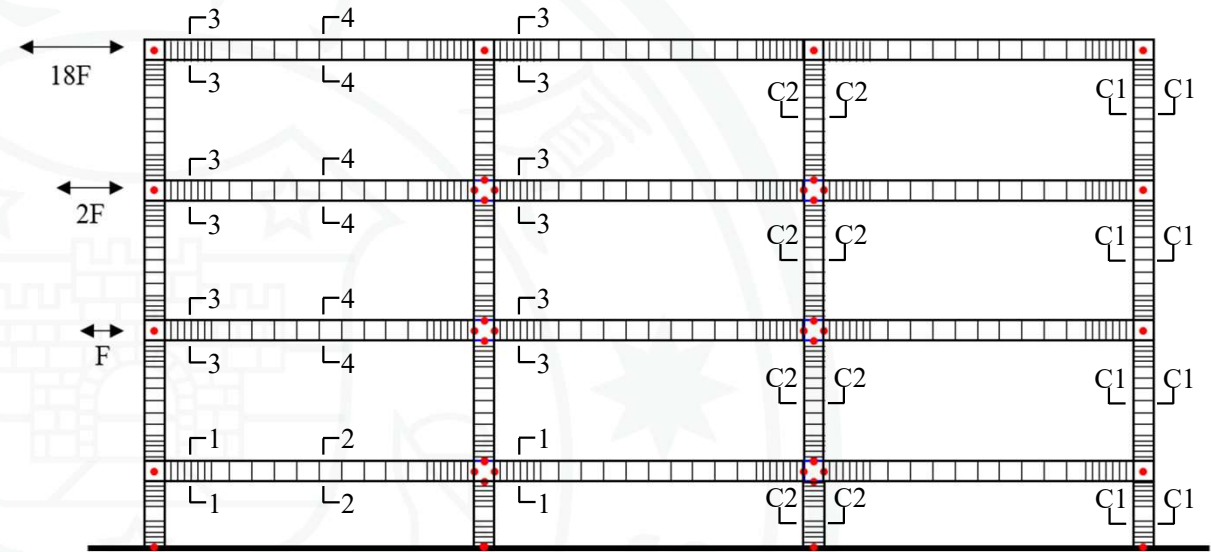
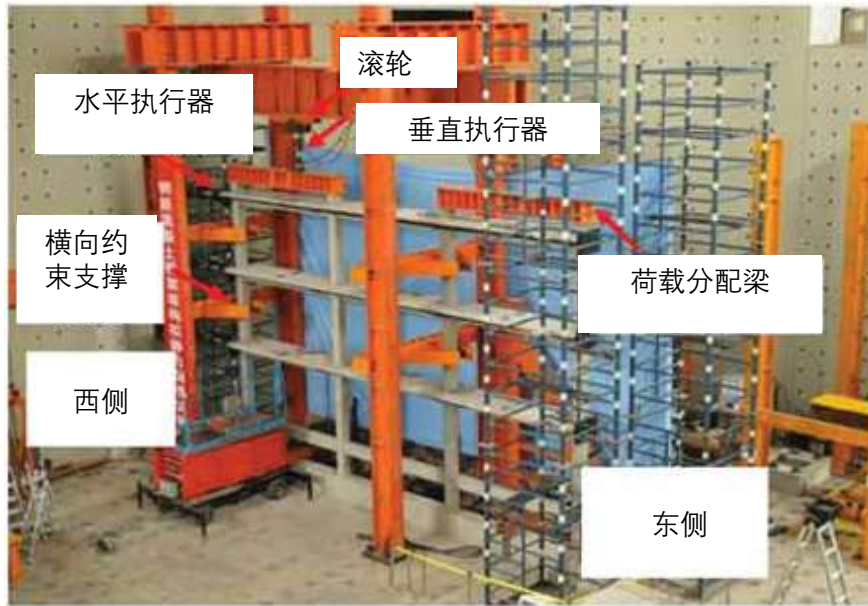
2. Opensees计算结果与试验数据对比



OpenSees 与试验滞回曲线总体吻合。正、负峰值承载力误差最大分别为 9.62%和 9.06%。大位移阶段模型强度退化略弱于试验。

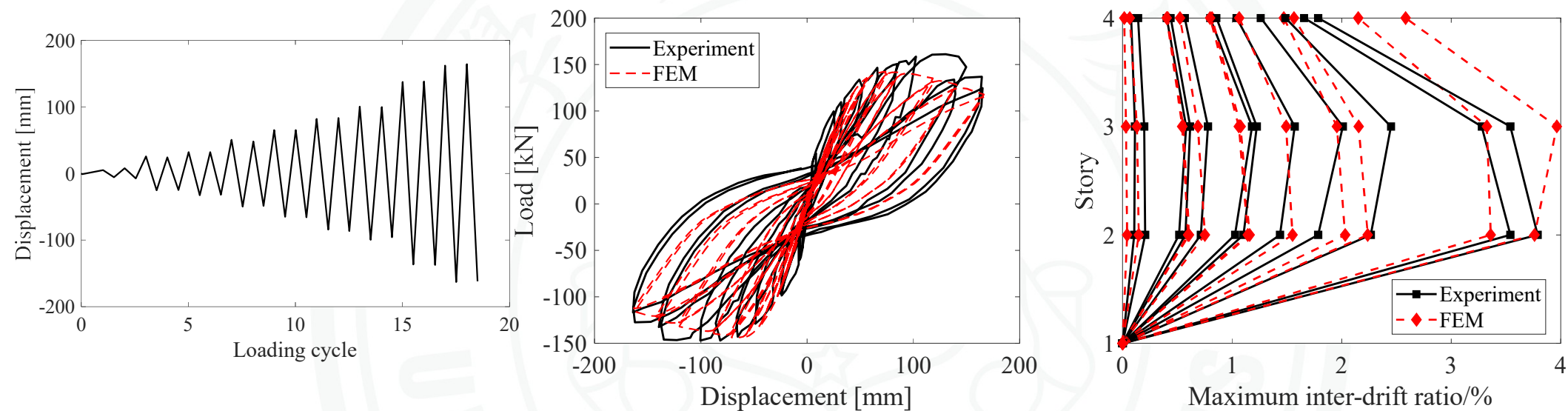
算例3：非线性RC框架结构振动台试验的
OpenSees数值模拟与试验结果对比

1. 结构概况



试验引用: L. Xie, X. Lu, H. Guan, and X. Lu, "Experimental study and numerical model calibration for earthquake-induced collapse of RC frames with emphasis on key columns, joints, and the overall structure," *Journal of Earthquake Engineering*, vol. 19, no. 8, pp. 1320-1344, 2015

1. Opensees计算结果与试验数据对比

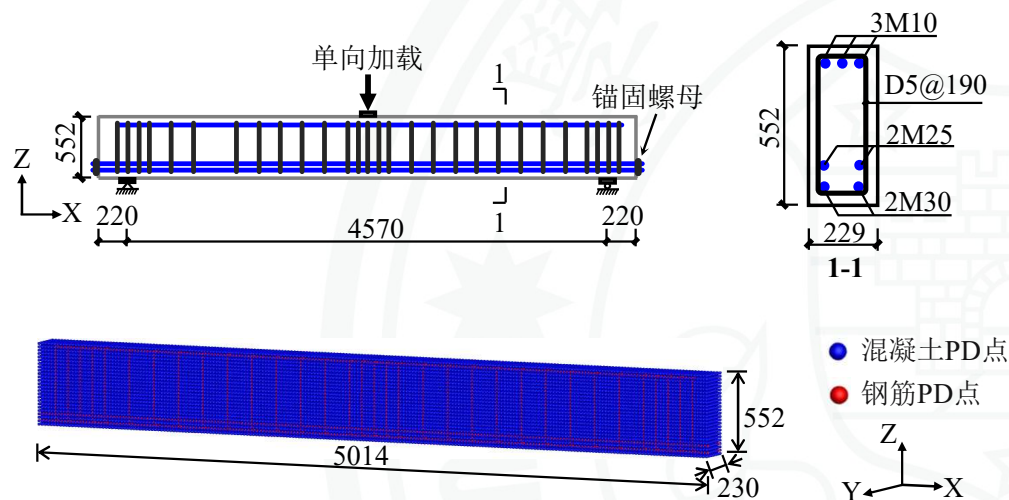


OpenSees与试验滞回曲线总体吻合。正、负峰值承载力误差最大分别为17.73%和17.57%，模型能够较好反映框架的滞回特征及层间变形分布，但正向峰值承载力略低于试验结果。

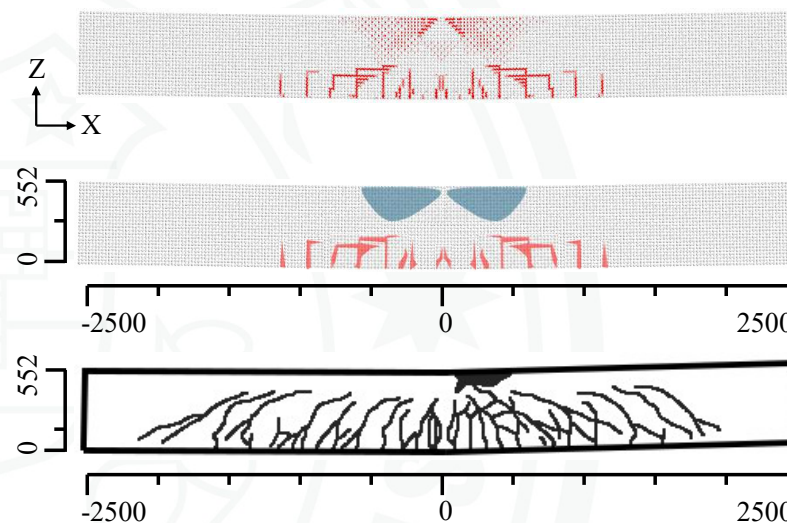
**算例4：非线性RC构件基于
Opensees的Peridynamics模拟与试验对比**

1.钢筋混凝土梁模拟

RC梁单向加载



最终破坏的PD模型损伤区域与试验对比

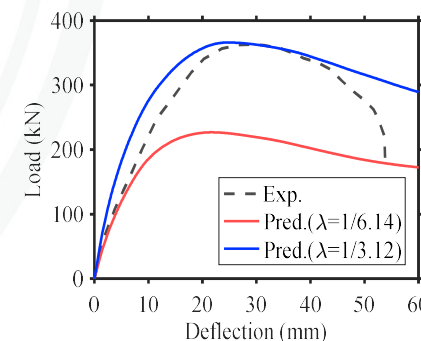


BPD 模拟结果与试验荷载-位移曲线总体吻合。采用 $\lambda=1/3.12$ 时，峰值承载力误差约为 4.17%；大位移阶段承载力退化趋势与试验基本一致。

试验引用：Shen X, Li B, Chen Y-T, Tizani W. Seismic performance of reinforced concrete ;interior beam-column joints with novel reinforcement detail. Eng Struct 2021;227: 111408.

Leon RT. Shear strength and hysteretic behavior of interior beam-column joints. Struct J 1990;87:3-11.

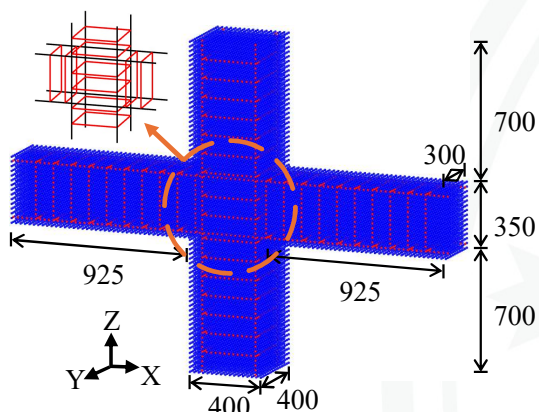
对比引用：Peridynamic modeling of RC beam-column joints under cyclic loading. Structures, 2025,73,108305.



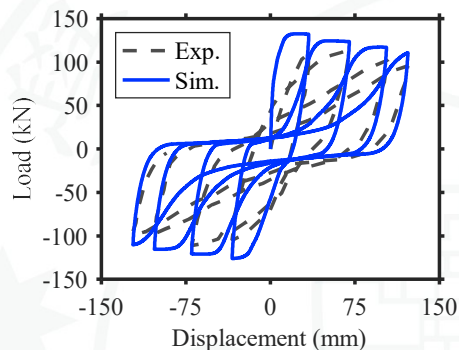
荷载与位移关系曲线

2.钢筋混凝土梁柱节点模拟

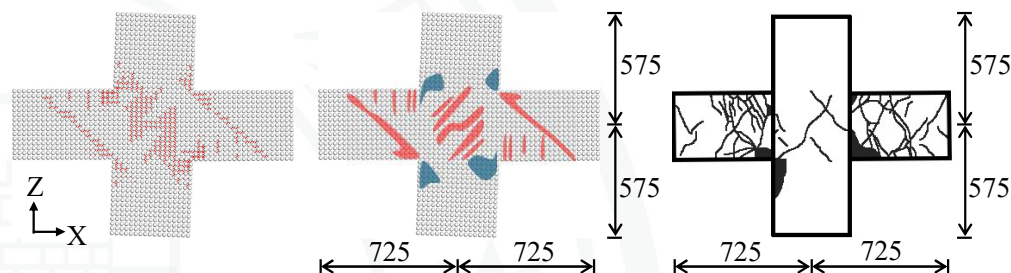
梁端开裂节点循环加载



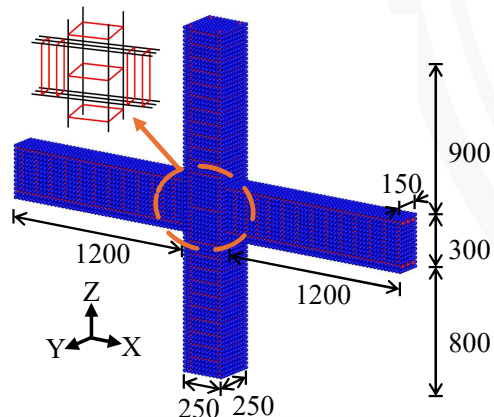
荷载与位移关系的滞回曲线



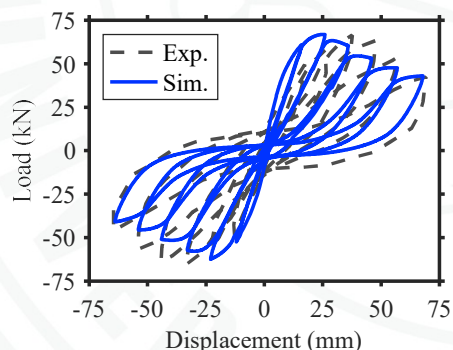
最终破坏的PD模型损伤区域与试验对比



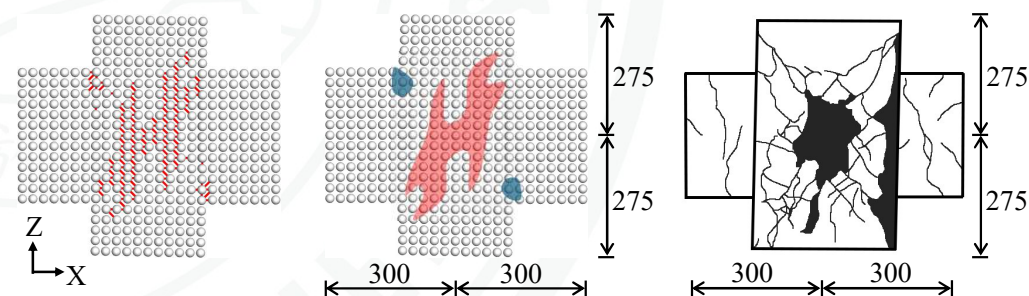
剪切破坏节点循环加载



荷载与位移关系的滞回曲线



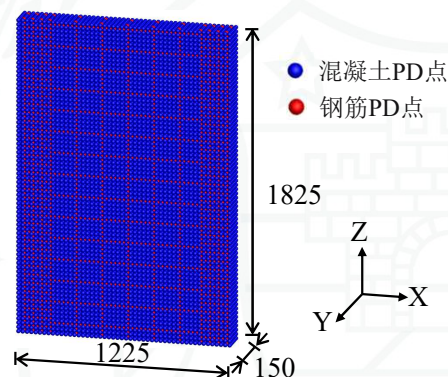
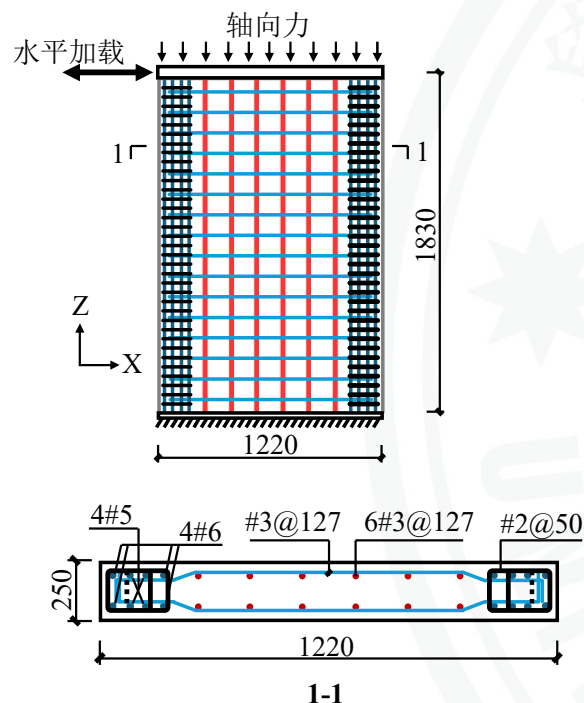
最终破坏的PD模型损伤区域与试验对比



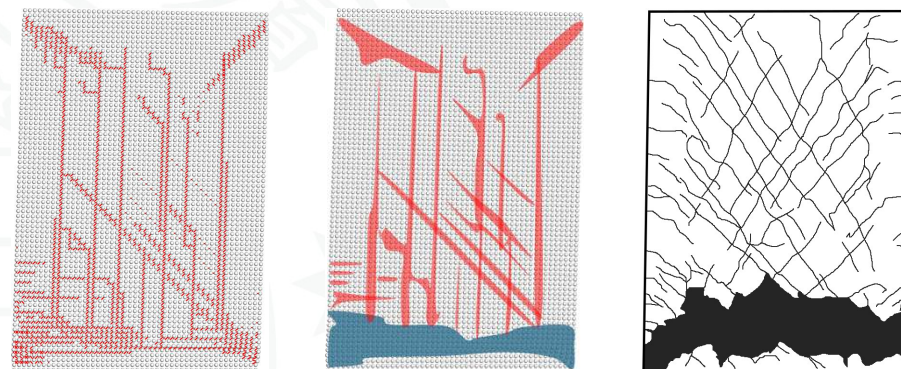
BPD 模型能够较好模拟两类梁柱节点的滞回响应和最终破坏形态。梁端开裂节点正、负峰值承载力误差分别为 12.4% 和 14.0%；剪切破坏节点正、负峰值承载力误差分别约为 8.3% 和 18.6%。模型能够较好反映节点区损伤集中和斜裂缝发展。

3.钢筋混凝土剪力墙模拟

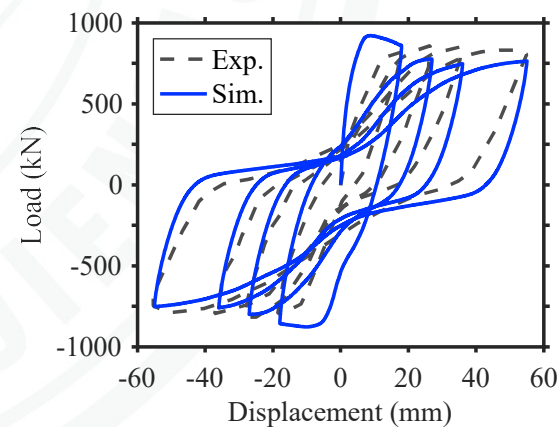
弯剪破坏剪力墙循环加载



最终破坏的PD模型损伤区域与试验对比



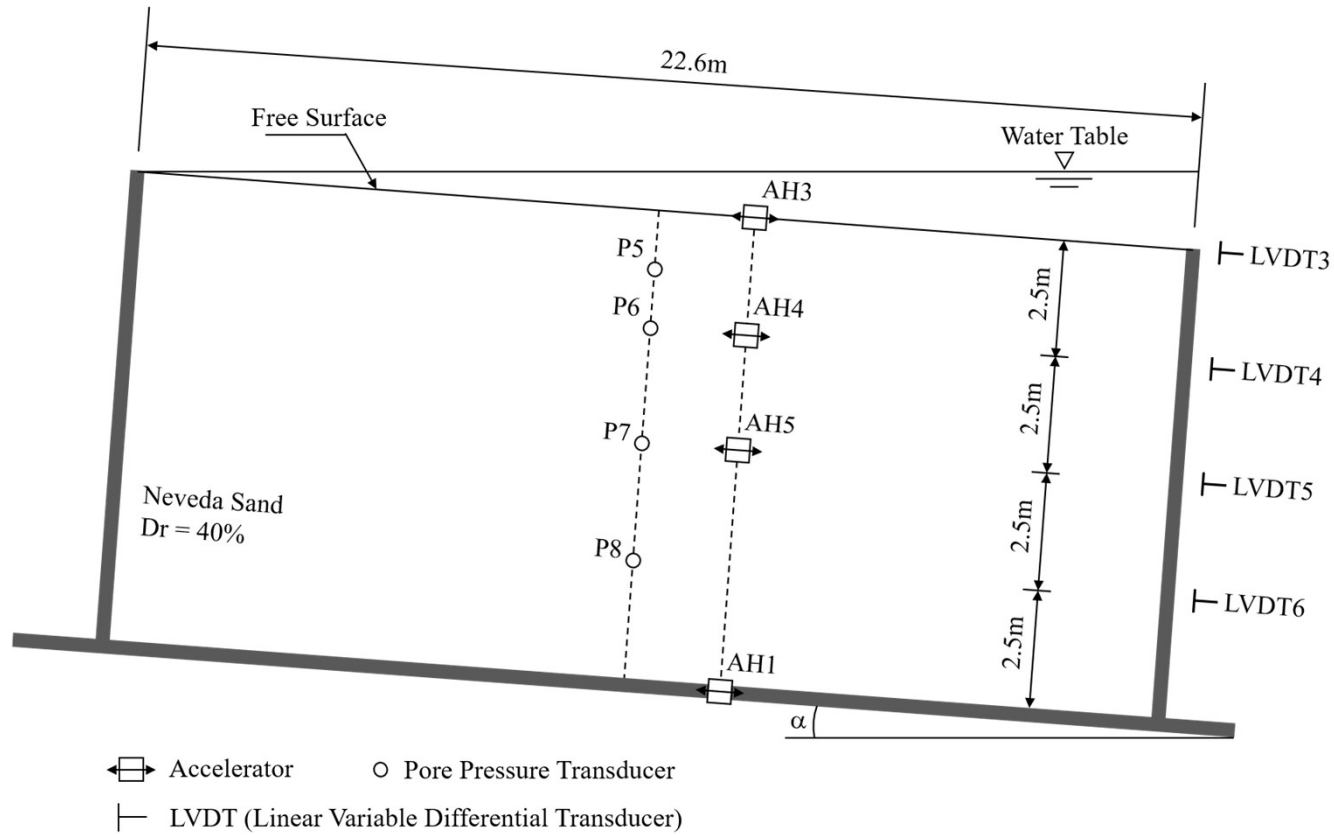
荷载与位移关系的滞回曲线



BPD 模拟结果能够较好反映剪力墙弯剪破坏下的滞回响应和损伤分布。正、负峰值承载力误差分别约为 8.4% 和 13.8%。模型能够捕捉墙底压碎、斜裂缝发展和边缘区域损伤，但滞回曲线较试验略饱满，对捏拢效应和耗能退化的模拟仍有一定偏差。

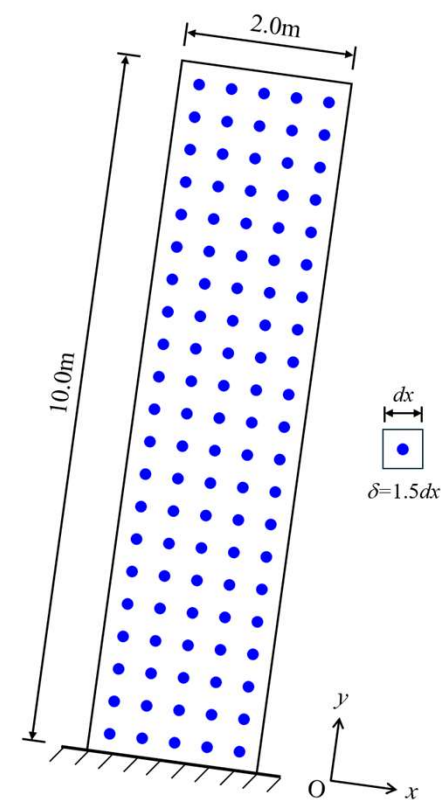
**算例5：非线性岩土边坡地震液化基于
Opensees的Peridynamics模拟与试验对比**

1. 试验模型与测点布置



试验引用: Taboada, V. Experimental results of model 2 at RPI. in Proc Int. Conf. Verification of Numerical Procedures for the Analysis of Soil Liquefaction Problems. 1993.

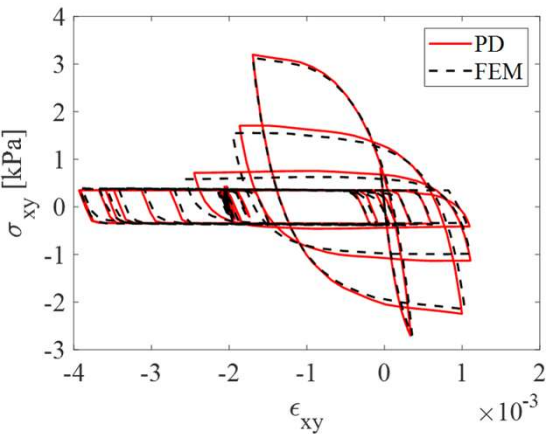
2. PD土柱模型与材料参数



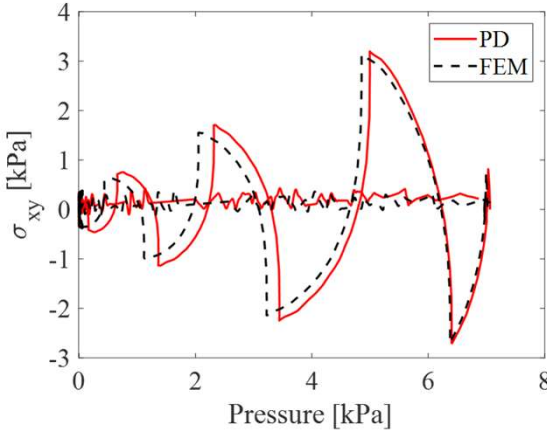
(a) Representative PD soil column model

(b) The parameters used in the PDMY material

Parameter	Value
Reference low-strain shear modulus G_r (kPa)	4.0×10^4
Reference bulk modulus B_r (kPa)	2.0×10^5
Friction angle at peak shear strength ϕ	30
Peak shear strain	0.1
Reference mean effective confining pressure (kPa)	80
Pressure depend coefficient d	0.5
Phase transformation angle ϕ_{PT}	28
Contraction	0.03
Dilation 1 and 2	0.4 and 1.0
Liquefaction parameter 1, 2 and 3	10, 0.02 and 1.0

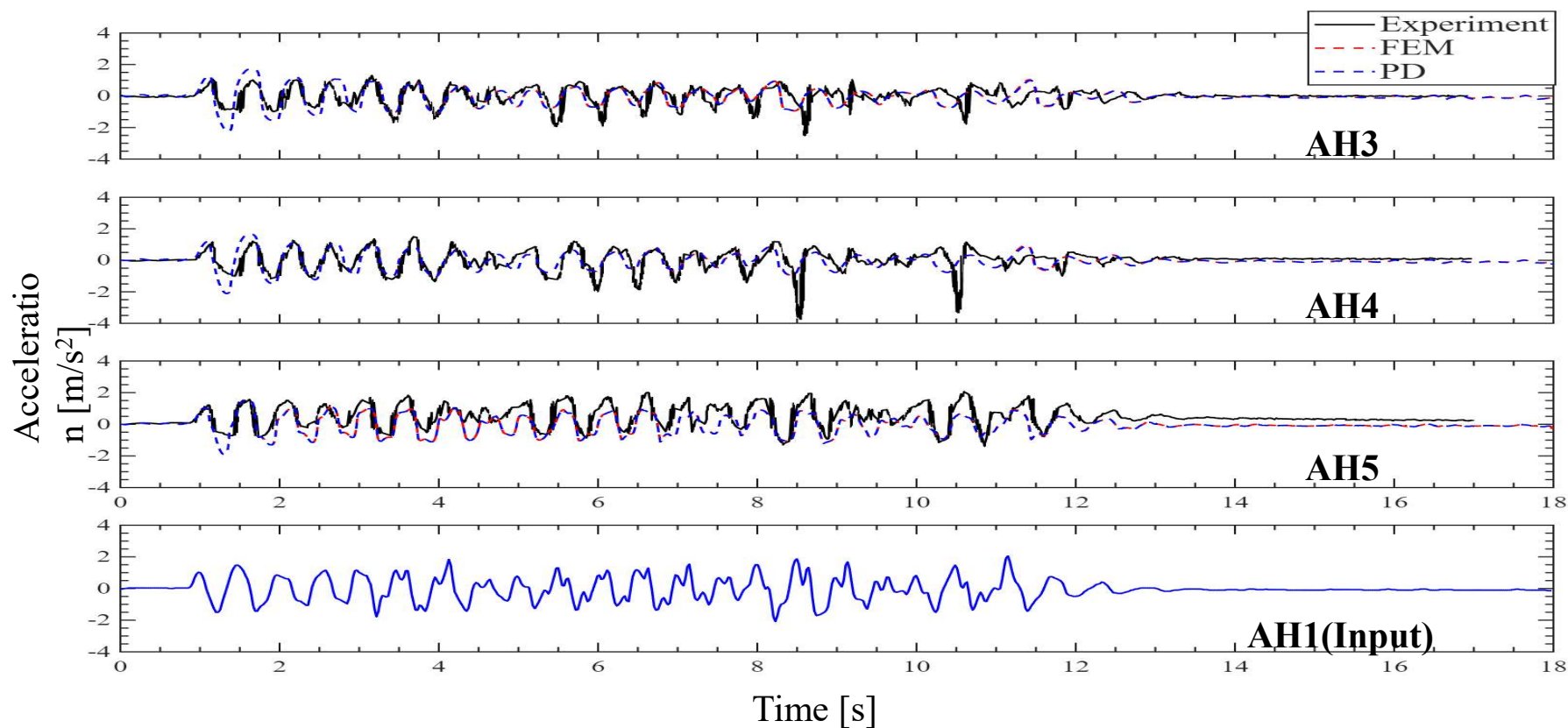


(a) shear stress vs. strain



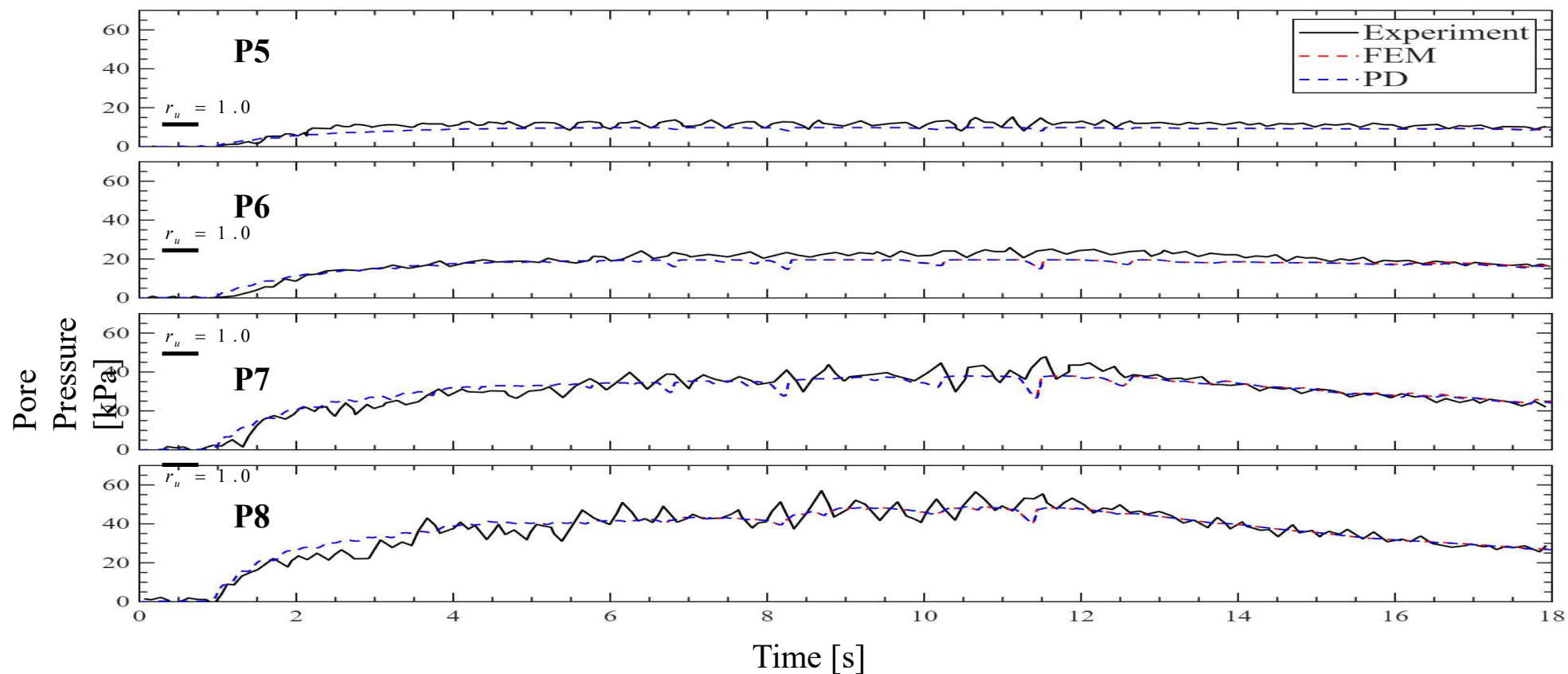
(b) shear stress vs. mean effective pressure

3. 加速度时程响应对比



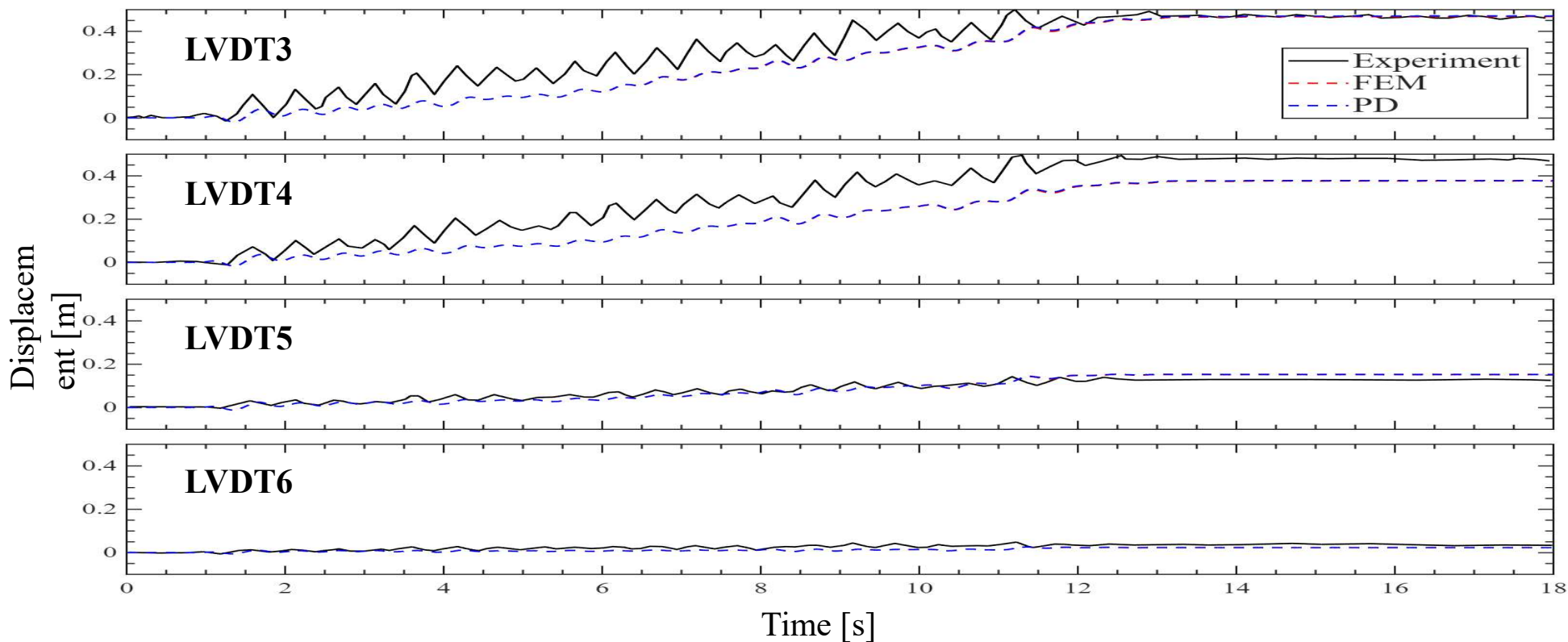
OpenSees-PD 模型能够基本反映不同高度测点的加速度时程变化趋势。AH3、AH4 和 AH5 的峰值加速度误差分别约为 24.0%、42.7% 和 28.3%，其中 AH4 测点偏差最大。模型对加速度响应的整体频率特征有一定模拟能力，但局部峰值和相位仍存在偏差。

4. 孔隙水压力时程响应对比



OpenSees-PD 模型能够较好模拟孔隙水压力的累积、增长和后期消散趋势。孔压峰值最大误差约为 23.2%。整体上，模型对液化过程中超孔压发展的模拟效果较好

5. 侧向位移时程响应对比



OpenSees-PD 模型能够反映边坡侧向位移逐步发展的总体趋势。主要位移测点的峰值位移最大误差约为 28.2%，主要出现在 LVDT4 测点。LVDT5 和 LVDT6 的位移幅值较小，相对误差受图中读数影响较大，不宜单独作为主要评价指标。